

AWS Lex Chatbot Creation Project – Implementation Report

Student

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Introduction

The purpose of this project was to design and implement a chatbot using Amazon Lex, a conversational AI service from Amazon Web Services (AWS). Chatbots allow users to interact with systems using natural language. In this project the chatbot was built to demonstrate how conversational interfaces work using intents, utterances and responses.

Objectives

- Understand chatbot development using Amazon Lex
- Create a simple chatbot capable of answering a predefined question
- Extend the chatbot with a quiz feature
- Demonstrate conversational logic using intents and utterances

Tools and Technologies

Amazon Web Services (AWS)
Amazon Lex
GitHub
Microsoft Word / PowerPoint

Architecture

User → Amazon Lex Bot → Intent Recognition → Response / Quiz Logic → Response to User

Part 1 – Basic Chatbot

A chatbot was created using Amazon Lex with the following components:

- One intent

- One utterance
- One predefined response

The bot was then tested in the Lex testing window to confirm the response was correct.

Part 2 – Quiz Chatbot

The chatbot was extended with a quiz feature to test users' knowledge of AWS services. Users can start the quiz, answer multiple-choice questions and receive feedback for correct or incorrect answers.

Testing and Results

The chatbot was tested using the Amazon Lex testing console. Testing confirmed that the chatbot correctly recognized user input and returned the expected responses.

Conclusion

This project demonstrated how Amazon Lex can be used to create conversational chatbots. The chatbot successfully answered AWS questions and provided an interactive quiz experience.